

# City Council Districts 1 - 7 Linear Regression Model Comparison: R | SPSS 2023 - 2025 Sales

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## Model I - Logarithm of Sale Price per Square Foot (LN\_SPPSF)

### Linear Regression Fit Statistics

Table 1: District 1 | LN\_SPPSF Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 2055     | 2041     | 14     | 0.4487 | 0.1150 | 0.1090 | 18.9110 |
| R                  | 2055     | 2041     | 14     | 0.4487 | 0.1148 | 0.1088 | 18.9112 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | -0.00% | -0.17% | -0.18% | 0.00%   |

Table 2: District 2 | LN\_SPPSF Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 325      | 319      | 6      | 0.4819 | 0.3140 | 0.3010 | 24.3360 |
| R                  | 325      | 319      | 6      | 0.4819 | 0.3140 | 0.3011 | 24.3362 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | -0.01% | 0.00%  | 0.03%  | 0.00%   |

Table 3: District 3 | LN\_SPPSF Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat |
|--------------------|----------|----------|--------|--------|--------|--------|--------|
| SPSS               | 949      | 938      | 11     | 0.4874 | 0.0950 | 0.0840 | 8.9520 |
| R                  | 949      | 938      | 11     | 0.4877 | 0.0939 | 0.0832 | 8.8333 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.06%  | -1.16% | -0.95% | -1.33% |

Table 4: District 4 | LN\_SPPSF Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1297     | 1286     | 11     | 0.4876 | 0.0990 | 0.0910 | 12.8330 |
| R                  | 1297     | 1286     | 11     | 0.4876 | 0.0989 | 0.0912 | 12.8330 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | -0.01% | -0.10% | 0.22%  | 0.00%   |

Table 5: District 5 | LN\_SPPSF Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 160      | 153      | 7      | 0.6275 | 0.3930 | 0.3650 | 14.1390 |
| R                  | 160      | 153      | 7      | 0.6275 | 0.3928 | 0.3650 | 14.1386 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.01%  | -0.05% | 0.00%  | -0.00%  |

Table 6: District 6 | LN\_SPPSF Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat |
|--------------------|----------|----------|--------|--------|--------|--------|--------|
| SPSS               | 193      | 182      | 11     | 0.5004 | 0.2070 | 0.1590 | 4.3120 |
| R                  | 193      | 181      | 12     | 0.5010 | 0.2094 | 0.1570 | 3.9961 |
| % diff (R vs SPSS) | 0.00%    | -0.55%   | 9.09%  | 0.11%  | 1.16%  | -1.26% | -7.33% |

Table 7: District 7 | LN\_SPPSF Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1385     | 1372     | 13     | 0.4507 | 0.1380 | 0.1300 | 16.9220 |
| R                  | 1385     | 1370     | 15     | 0.4521 | 0.1341 | 0.1247 | 14.1499 |
| % diff (R vs SPSS) | 0.00%    | -0.15%   | 15.38% | 0.30%  | -2.83% | -4.08% | -16.38% |

## Model II - Logarithm of Sale Price (LN\_PRICE) excluding ECFs

### Linear Regression Fit Statistics

Table 8: District 1 | LN\_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 2055     | 2035     | 20     | 0.4459 | 0.3500 | 0.3440 | 54.8720 |
| R                  | 2055     | 2035     | 20     | 0.4459 | 0.3503 | 0.3440 | 54.8718 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | -0.01% | 0.09%  | 0.00%  | -0.00%  |

Table 9: District 2 | LN\_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1739     | 1721     | 18     | 0.4506 | 0.4850 | 0.4790 | 89.9480 |
| R                  | 1739     | 1721     | 18     | 0.4506 | 0.4847 | 0.4794 | 89.9476 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.01%  | -0.06% | 0.08%  | -0.00%  |

Table 10: District 3 | LN\_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1393     | 1375     | 18     | 0.5138 | 0.1570 | 0.1460 | 14.2420 |
| R                  | 1393     | 1375     | 18     | 0.5138 | 0.1570 | 0.1460 | 14.2306 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | -0.00% | 0.00%  | 0.00%  | -0.08%  |

Table 11: District 4 | LN\_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1707     | 1688     | 19     | 0.5020 | 0.2710 | 0.2620 | 32.9650 |
| R                  | 1707     | 1688     | 19     | 0.5020 | 0.2706 | 0.2624 | 32.9646 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.01%  | -0.15% | 0.15%  | -0.00%  |

Table 12: District 5 | LN\_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 307      | 289      | 18     | 0.6744 | 0.5350 | 0.5060 | 18.4470 |
| R                  | 307      | 289      | 18     | 0.6744 | 0.5346 | 0.5057 | 18.4465 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.01%  | -0.07% | -0.06% | -0.00%  |

Table 13: District 6 | LN\_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 299      | 285      | 14     | 0.5507 | 0.4550 | 0.4290 | 17.0240 |
| R                  | 298      | 283      | 15     | 0.5483 | 0.4415 | 0.4118 | 14.9114 |
| % diff (R vs SPSS) | -0.33%   | -0.70%   | 7.14%  | -0.44% | -2.97% | -4.01% | -12.41% |

Table 14: District 7 | LN\_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1805     | 1790     | 15     | 0.4652 | 0.2250 | 0.2190 | 34.6630 |
| R                  | 1805     | 1790     | 15     | 0.4680 | 0.2159 | 0.2094 | 32.8677 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.59%  | -4.04% | -4.38% | -5.18%  |

## Model III - Logarithm of Sale Price (LN\_PRICE)

### Linear Regression Fit Statistics

Table 15: District 1 | LN\_PRICE Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 2055     | 2017     | 38     | 0.4107 | 0.4540 | 0.4440 | 44.1250 |
| R                  | 2055     | 2017     | 38     | 0.4107 | 0.4539 | 0.4437 | 44.1245 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.01%  | -0.02% | -0.07% | -0.00%  |

Table 16: District 2 | LN\_PRICE Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1739     | 1707     | 32     | 0.4157 | 0.5650 | 0.5570 | 69.2910 |
| R                  | 1739     | 1707     | 32     | 0.4157 | 0.5650 | 0.5569 | 69.2908 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.00%  | 0.00%  | -0.02% | -0.00%  |

Table 17: District 3 | LN\_PRICE Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1393     | 1364     | 29     | 0.4650 | 0.3150 | 0.3010 | 21.6620 |
| R                  | 1393     | 1364     | 29     | 0.4651 | 0.3149 | 0.3003 | 21.6198 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.03%  | -0.03% | -0.23% | -0.19%  |

Table 18: District 4 | LN\_PRICE Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1707     | 1676     | 31     | 0.4533 | 0.4090 | 0.3980 | 37.4710 |
| R                  | 1707     | 1676     | 31     | 0.4533 | 0.4094 | 0.3985 | 37.4794 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | -0.00% | 0.10%  | 0.13%  | 0.02%   |

Table 19: District 5 | LN\_PRICE Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 307      | 281      | 26     | 0.5948 | 0.6480 | 0.6150 | 19.8960 |
| R                  | 307      | 281      | 26     | 0.5948 | 0.6480 | 0.6154 | 19.8957 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.00%  | 0.00%  | 0.07%  | -0.00%  |

Table 20: District 6 | LN\_PRICE Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 299      | 276      | 23     | 0.5072 | 0.5530 | 0.5150 | 14.8270 |
| R                  | 298      | 273      | 25     | 0.5050 | 0.5430 | 0.5011 | 12.9745 |
| % diff (R vs SPSS) | -0.33%   | -1.09%   | 8.70%  | -0.43% | -1.81% | -2.70% | -12.49% |

Table 21: District 7 | LN\_PRICE Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1805     | 1779     | 26     | 0.4302 | 0.3420 | 0.3320 | 35.4870 |
| R                  | 1805     | 1779     | 26     | 0.4317 | 0.3368 | 0.3272 | 34.7546 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.35%  | -1.52% | -1.45% | -2.06%  |

## Model IV - Logarithm of Sale Price (LN\_PRICE) with Outlier Removal

### Linear Regression Fit Statistics

Table 22: District 1 | LN\_PRICE (Inliers) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1898     | 1866     | 32     | 0.3676 | 0.4950 | 0.4870 | 57.2650 |
| R                  | 1898     | 1866     | 32     | 0.3676 | 0.4955 | 0.4868 | 57.2647 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.00%  | 0.10%  | -0.04% | -0.00%  |

Table 23: District 2 | LN\_PRICE (Inliers) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1593     | 1566     | 27     | 0.3745 | 0.5840 | 0.5770 | 81.4850 |
| R                  | 1593     | 1566     | 27     | 0.3745 | 0.5842 | 0.5770 | 81.4846 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.00%  | 0.03%  | 0.00%  | -0.00%  |

Table 24: District 3 | LN\_PRICE (Inliers) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1259     | 1232     | 27     | 0.3896 | 0.4010 | 0.3880 | 30.5860 |
| R                  | 1254     | 1226     | 28     | 0.3866 | 0.4015 | 0.3879 | 29.3791 |
| % diff (R vs SPSS) | -0.40%   | -0.49%   | 3.70%  | -0.77% | 0.12%  | -0.03% | -3.95%  |

Table 25: District 4 | LN\_PRICE (Inliers) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1576     | 1547     | 29     | 0.3945 | 0.4810 | 0.4720 | 49.5060 |
| R                  | 1576     | 1547     | 29     | 0.3945 | 0.4814 | 0.4717 | 49.5152 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | -0.00% | 0.08%  | -0.06% | 0.02%   |

Table 26: District 5 | LN\_PRICE (Inliers) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 262      | 238      | 24     | 0.4466 | 0.7790 | 0.7570 | 35.0370 |
| R                  | 260      | 237      | 23     | 0.4452 | 0.7767 | 0.7550 | 35.8318 |
| % diff (R vs SPSS) | -0.76%   | -0.42%   | -4.17% | -0.32% | -0.30% | -0.26% | 2.27%   |

Table 27: District 6 | LN\_PRICE (Inliers) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 244      | 224      | 20     | 0.3914 | 0.6350 | 0.6020 | 19.4500 |
| R                  | 246      | 225      | 21     | 0.3950 | 0.6423 | 0.6090 | 19.2430 |
| % diff (R vs SPSS) | 0.82%    | 0.45%    | 5.00%  | 0.92%  | 1.15%  | 1.16%  | -1.06%  |

Table 28: District 7 | LN\_PRICE (Inliers) Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1623     | 1598     | 25     | 0.3668 | 0.4260 | 0.4170 | 47.4560 |
| R                  | 1647     | 1621     | 26     | 0.3748 | 0.3936 | 0.3839 | 40.4716 |
| % diff (R vs SPSS) | 1.48%    | 1.44%    | 4.00%  | 2.18%  | -7.61% | -7.94% | -14.72% |

## Model V - Logarithm of Time Adjusted Sale Price (LN\_TASP)

### Linear Regression Fit Statistics

Table 29: District 1 | LN\_TASP Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1898     | 1867     | 31     | 0.3675 | 0.4870 | 0.4780 | 57.1660 |
| R                  | 1898     | 1867     | 31     | 0.3675 | 0.4870 | 0.4785 | 57.1664 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.00%  | 0.00%  | 0.10%  | 0.00%   |

Table 30: District 2 | LN\_TASP Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1593     | 1567     | 26     | 0.3744 | 0.5750 | 0.5680 | 81.6790 |
| R                  | 1593     | 1567     | 26     | 0.3898 | 0.5554 | 0.5481 | 75.3011 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 4.12%  | -3.41% | -3.50% | -7.81%  |

Table 31: District 3 | LN\_TASP Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1259     | 1234     | 25     | 0.3893 | 0.3920 | 0.3790 | 31.7870 |
| R                  | 1254     | 1228     | 26     | 0.3871 | 0.3923 | 0.3795 | 30.4959 |
| % diff (R vs SPSS) | -0.40%   | -0.49%   | 4.00%  | -0.57% | 0.08%  | 0.13%  | -4.06%  |

Table 32: District 4 | LN\_TASP Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1576     | 1548     | 28     | 0.3944 | 0.4810 | 0.4720 | 51.2550 |
| R                  | 1576     | 1548     | 28     | 0.3944 | 0.4810 | 0.4717 | 51.2463 |
| % diff (R vs SPSS) | 0.00%    | 0.00%    | 0.00%  | 0.01%  | 0.00%  | -0.06% | -0.02%  |

Table 33: District 5 | LN\_TASP Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 262      | 239      | 23     | 0.4458 | 0.7740 | 0.7520 | 35.5210 |
| R                  | 260      | 239      | 21     | 0.4473 | 0.7671 | 0.7466 | 37.4852 |
| % diff (R vs SPSS) | -0.76%   | 0.00%    | -8.70% | 0.35%  | -0.89% | -0.72% | 5.53%   |

Table 34: District 6 | LN\_TASP Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 244      | 225      | 19     | 0.3905 | 0.6350 | 0.6040 | 20.5970 |
| R                  | 246      | 225      | 21     | 0.3962 | 0.6402 | 0.6066 | 19.0636 |
| % diff (R vs SPSS) | 0.82%    | 0.00%    | 10.53% | 1.45%  | 0.82%  | 0.43%  | -7.44%  |

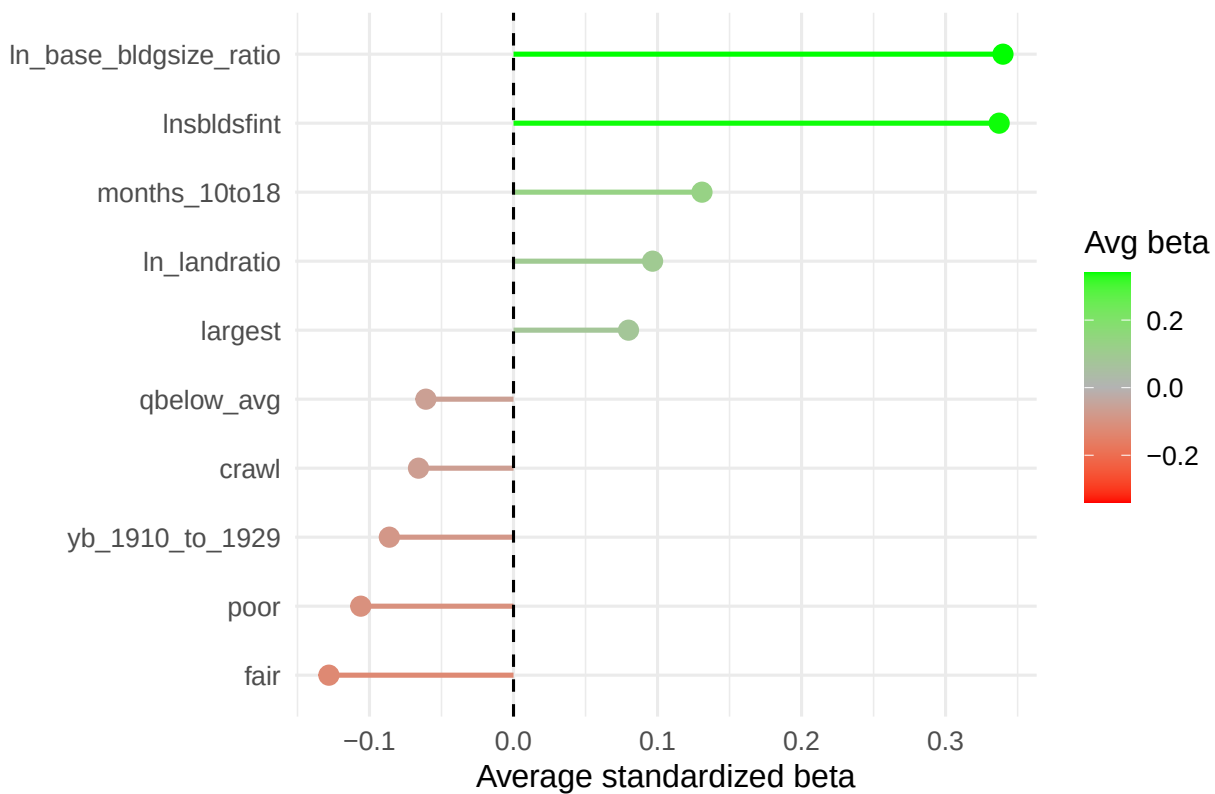
Table 35: District 7 | LN\_TASP Model Fit Statistics: SPSS vs R

| metricsource       | df_total | df_resid | df_reg | see    | r2     | adj_r2 | f_stat  |
|--------------------|----------|----------|--------|--------|--------|--------|---------|
| SPSS               | 1623     | 1599     | 24     | 0.3667 | 0.4020 | 0.3930 | 44.7940 |
| R                  | 1647     | 1622     | 25     | 0.3751 | 0.3676 | 0.3578 | 37.7124 |
| % diff (R vs SPSS) | 1.48%    | 1.44%    | 4.17%  | 2.30%  | -8.56% | -8.96% | -15.81% |

## Predictor Impact by District

### District 1

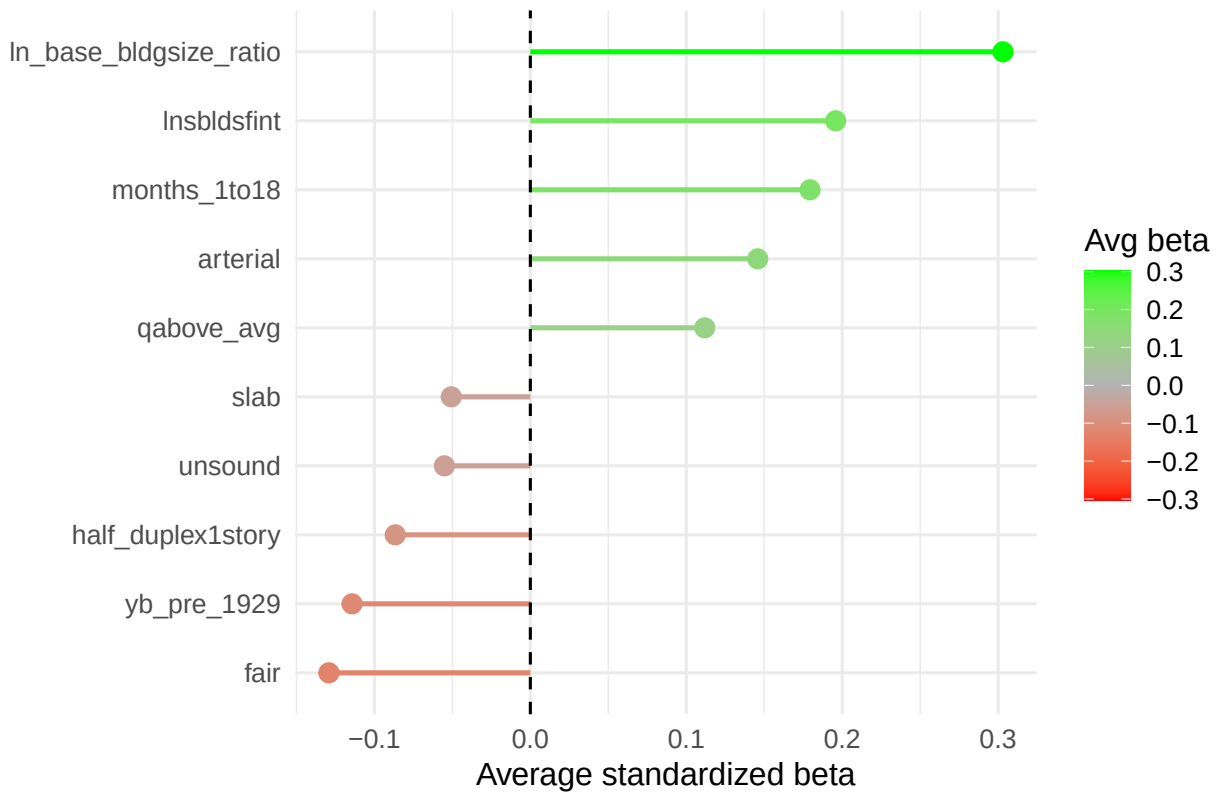
#### District 1 | Average Standardized Beta across Models



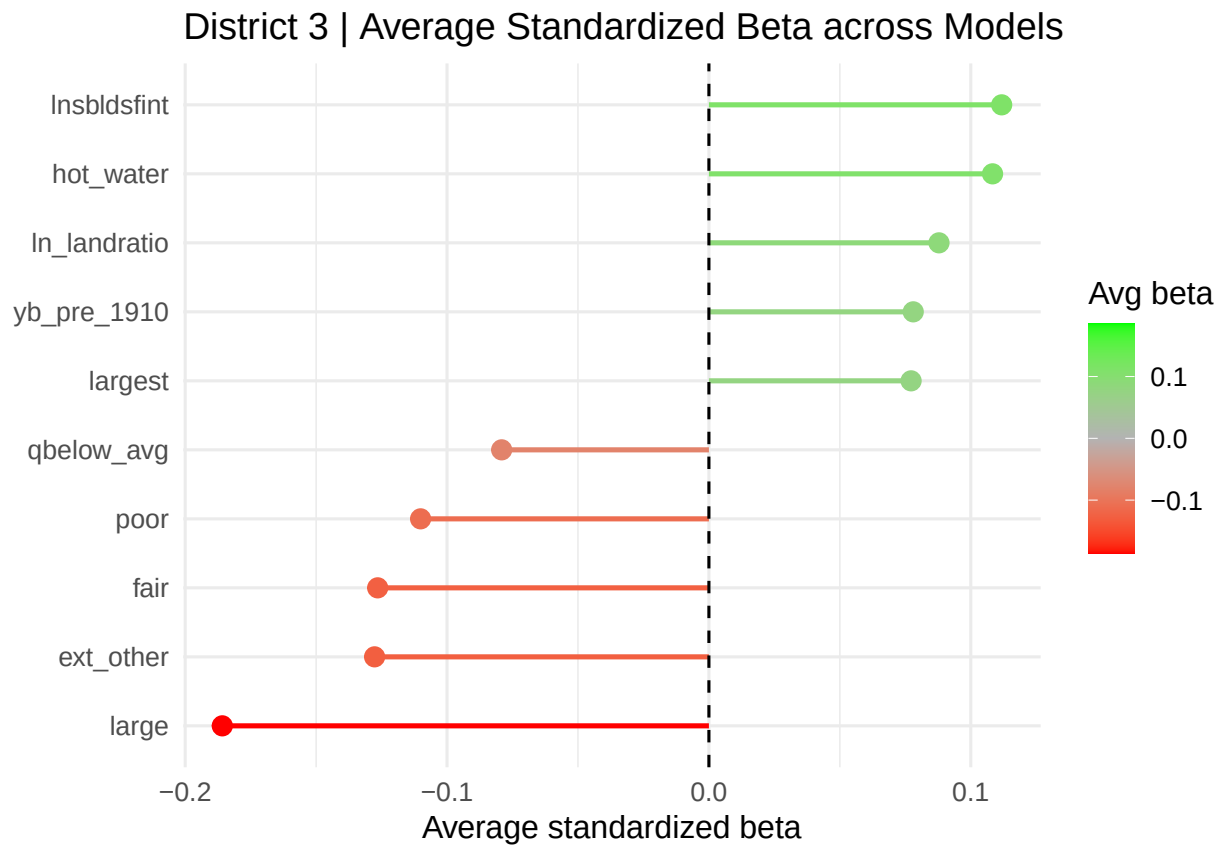


## District 2

### District 2 | Average Standardized Beta across Models

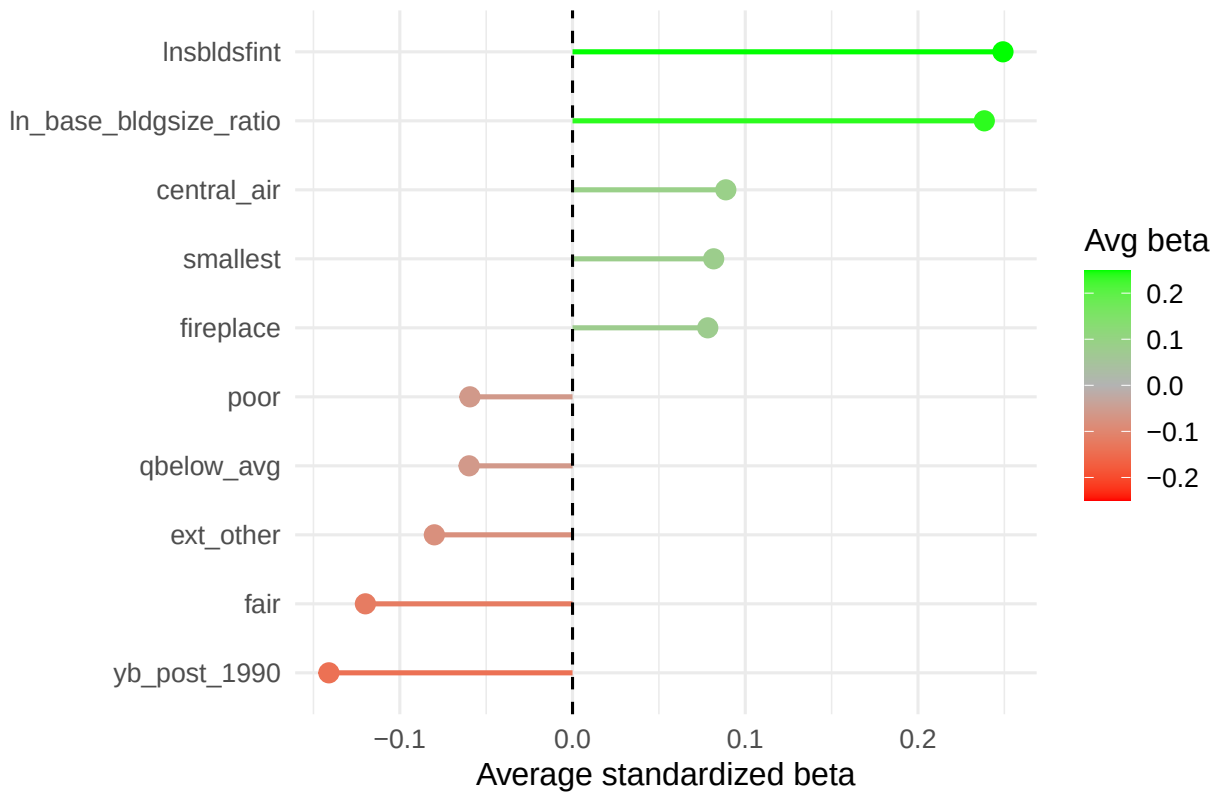


### District 3



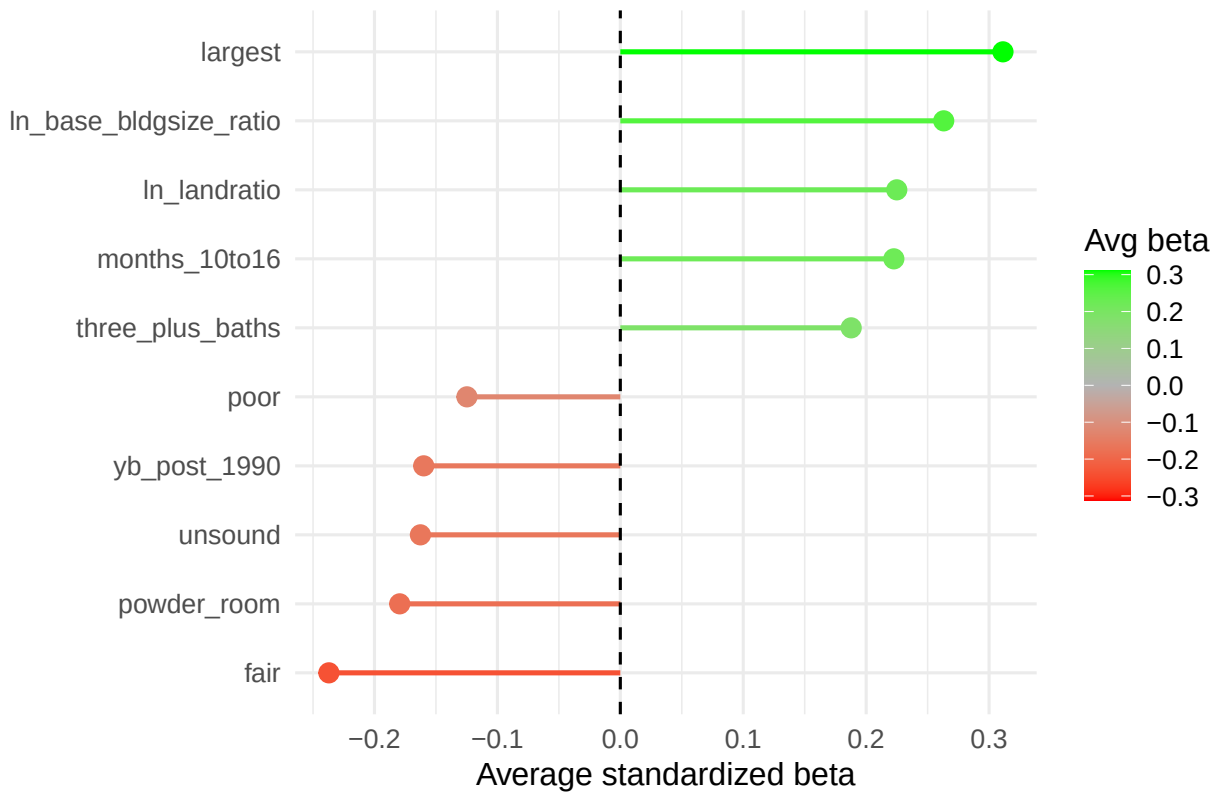
## District 4

### District 4 | Average Standardized Beta across Models



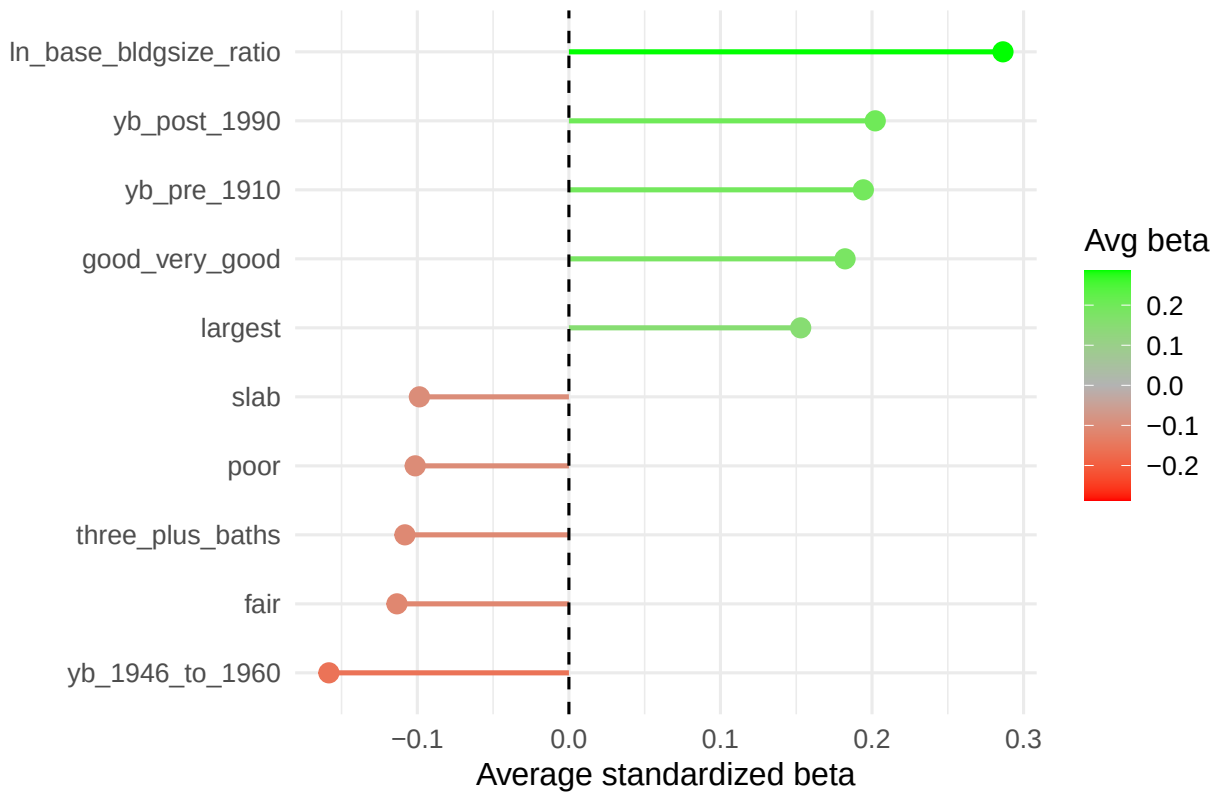
## District 5

### District 5 | Average Standardized Beta across Models



## District 6

### District 6 | Average Standardized Beta across Models



## District 7

### District 7 | Average Standardized Beta across Models

